

Step 1: Importing Python Libraries

```
In [1]: # Importing libraries

import pandas as pd
import numpy as np
import os
from datetime import datetime
```

Step 2: Importing Stock Datasets

```
In [2]: # Importing Amazon stock data

path = r'/Users/matthewabrams/Desktop/Stocks Data/Stock Datasets/'
amazon = pd.read_csv(os.path.join(path, 'Original Stock Data/HistoricalData_Amaz
amazon
```

Out[2]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	2015-01-02	15.629000	15.737500	15.348000	15.426000	15.426000	55664000
1	2015-01-05	15.350500	15.419000	15.042500	15.109500	15.109500	55484000
2	2015-01-06	15.112000	15.150000	14.619000	14.764500	14.764500	70380000
3	2015-01-07	14.875000	15.064000	14.766500	14.921000	14.921000	52806000
4	2015-01-08	15.016000	15.157000	14.805500	15.023000	15.023000	61768000
...	...	...	...	...	...	...	...
2511	2024-12-24	226.940002	229.139999	226.130005	229.050003	229.050003	15007500
2512	2024-12-26	228.500000	228.500000	226.669998	227.050003	227.050003	16146700
2513	2024-12-27	225.600006	226.029999	220.899994	223.750000	223.750000	27367100
2514	2024-12-30	220.059998	223.000000	218.429993	221.300003	221.300003	28321200
2515	2024-12-31	222.970001	223.229996	218.940002	219.389999	219.389999	24819700

2516 rows x 7 columns

```
In [3]: # Importing Apple stock data

apple = pd.read_csv(os.path.join(path, 'Original Stock Data/HistoricalData_Apple
apple
```

Out[3]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	2015-01-02	27.847500	27.860001	26.837500	27.332500	24.347174	212818400
1	2015-01-05	27.072500	27.162500	26.352501	26.562500	23.661274	257142000
2	2015-01-06	26.635000	26.857500	26.157499	26.565001	23.663502	263188400
3	2015-01-07	26.799999	27.049999	26.674999	26.937500	23.995319	160423600
4	2015-01-08	27.307501	28.037500	27.174999	27.972500	24.917271	237458000

	Date	Open	High	Low	Close	Adj Close	Volume
...	...	...	...	...	...	...	...
2511	2024-12-24	255.490005	258.209991	255.289993	258.200012	258.200012	23234700
2512	2024-12-26	258.190002	260.100006	257.630005	259.019989	259.019989	27237100
2513	2024-12-27	257.829987	258.700012	253.059998	255.589996	255.589996	42355300
2514	2024-12-30	252.229996	253.500000	250.750000	252.199997	252.199997	35557500
2515	2024-12-31	252.440002	253.279999	249.429993	250.419998	250.419998	39480700

2516 rows x 7 columns

In [4]:

```
# Importing Meta stock data

meta = pd.read_csv(os.path.join(path, 'Original Stock Data/HistoricalData_Meta.c
meta
```

Out[4]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	2015-01-02	78.580002	78.930000	77.699997	78.449997	78.151466	18177500
1	2015-01-05	77.980003	79.250000	76.860001	77.190002	76.896263	26452200
2	2015-01-06	77.230003	77.589996	75.360001	76.150002	75.860222	27399300
3	2015-01-07	76.760002	77.360001	75.820000	76.150002	75.860222	22045300
4	2015-01-08	76.739998	78.230003	76.080002	78.180000	77.882500	23961000
...	...	...	...	...	...	...	...
2511	2024-12-24	602.719971	607.989990	599.280029	607.750000	607.750000	4726100
2512	2024-12-26	605.479980	606.299988	598.940002	603.349976	603.349976	6081400
2513	2024-12-27	599.409973	601.849976	589.799988	599.809998	599.809998	8084200
2514	2024-12-30	588.750000	596.940002	585.580017	591.239990	591.239990	7025900
2515	2024-12-31	592.270020	593.969971	583.849976	585.510010	585.510010	6019500

2516 rows x 7 columns

In [5]:

```
# Importing Microsoft stock data

microsoft = pd.read_csv(os.path.join(path, 'Original Stock Data/HistoricalData_M
microsoft
```

Out[5]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	2015-01-02	46.660000	47.419998	46.540001	46.759998	40.152489	27913900
1	2015-01-05	46.369999	46.730000	46.250000	46.330002	39.783257	39673900
2	2015-01-06	46.380001	46.750000	45.540001	45.650002	39.199333	36447900
3	2015-01-07	45.980000	46.459999	45.490002	46.230000	39.697376	29114100

	Date	Open	High	Low	Close	Adj Close	Volume
4	2015-01-08	46.750000	47.750000	46.720001	47.590000	40.865200	29645200
...	...	...	...	...	...	...	...
2511	2024-12-24	434.649994	439.600006	434.190002	439.329987	439.329987	7164500
2512	2024-12-26	439.079987	440.940002	436.630005	438.109985	438.109985	8194200
2513	2024-12-27	434.600006	435.220001	426.350006	430.529999	430.529999	18117700
2514	2024-12-30	426.059998	427.549988	421.899994	424.829987	424.829987	13158700
2515	2024-12-31	426.100006	426.730011	420.660004	421.500000	421.500000	13246500

2516 rows x 7 columns

In [6]:

```
# Importing Netflix stock data

netflix = pd.read_csv(os.path.join(path, 'Original Stock Data/HistoricalData_Netflix'))
```

Out[6]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	2015-01-02	49.151428	50.331429	48.731430	49.848572	49.848572	13475000
1	2015-01-05	49.258572	49.258572	47.147144	47.311428	47.311428	18165000
2	2015-01-06	47.347141	47.639999	45.661430	46.501431	46.501431	16037700
3	2015-01-07	47.347141	47.421429	46.271427	46.742859	46.742859	9849700
4	2015-01-08	47.119999	47.835712	46.478573	47.779999	47.779999	9601900
...	...	...	...	...	...	...	...
2511	2024-12-24	915.000000	935.849976	911.700012	932.119995	932.119995	2320300
2512	2024-12-26	928.400024	930.489990	915.299988	924.140015	924.140015	2340300
2513	2024-12-27	916.010010	918.130005	894.500000	907.549988	907.549988	3226200
2514	2024-12-30	894.510010	908.229980	889.710022	900.429993	900.429993	2203000
2515	2024-12-31	901.799988	902.679993	889.469971	891.320007	891.320007	1875900

2516 rows x 7 columns

In [7]:

```
# Importing Tesla stock data

tesla = pd.read_csv(os.path.join(path, 'Original Stock Data/HistoricalData_Tesla'))
```

Out[7]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	2015-01-02	14.858000	14.883333	14.217333	14.620667	14.620667	71466000
1	2015-01-05	14.303333	14.433333	13.810667	14.006000	14.006000	80527500
2	2015-01-06	14.004000	14.280000	13.614000	14.085333	14.085333	93928500

	Date	Open	High	Low	Close	Adj Close	Volume
3	2015-01-07	14.223333	14.318667	13.985333	14.063333	14.063333	44526000
4	2015-01-08	14.187333	14.253333	14.000667	14.041333	14.041333	51637500
...	...	...	...	...	...	...	...
2511	2024-12-24	435.899994	462.779999	435.140015	462.279999	462.279999	59551800
2512	2024-12-26	465.160004	465.329987	451.019989	454.130005	454.130005	76366400
2513	2024-12-27	449.519989	450.000000	426.500000	431.660004	431.660004	82666800
2514	2024-12-30	419.399994	427.000000	415.750000	417.410004	417.410004	64941000
2515	2024-12-31	423.790009	427.929993	402.540009	403.839996	403.839996	76825100

2516 rows x 7 columns

Step 3: Observing Datatypes for all datasets

```
In [8]: amazon.dtypes
```

```
Out[8]: Date          object
Open          float64
High          float64
Low           float64
Close         float64
Adj Close     float64
Volume        int64
dtype: object
```

```
In [9]: apple.dtypes
```

```
Out[9]: Date          object
Open          float64
High          float64
Low           float64
Close         float64
Adj Close     float64
Volume        int64
dtype: object
```

```
In [10]: meta.dtypes
```

```
Out[10]: Date          object
Open          float64
High          float64
Low           float64
Close         float64
Adj Close     float64
Volume        int64
dtype: object
```

```
In [11]: microsoft.dtypes
```

```
Out[11]: Date          object
Open          float64
```

```
High          float64
Low           float64
Close         float64
Adj Close     float64
Volume        int64
dtype: object
```

```
In [12]: netflix.dtypes
```

```
Out[12]: Date          object
Open          float64
High          float64
Low           float64
Close         float64
Adj Close     float64
Volume        int64
dtype: object
```

```
In [13]: tesla.dtypes
```

```
Out[13]: Date          object
Open          float64
High          float64
Low           float64
Close         float64
Adj Close     float64
Volume        int64
dtype: object
```

Step 4: Checking for missing values

```
In [14]: amazon.isnull().sum()
```

```
Out[14]: Date          0
Open          0
High          0
Low           0
Close         0
Adj Close     0
Volume        0
dtype: int64
```

```
In [15]: apple.isnull().sum()
```

```
Out[15]: Date          0
Open          0
High          0
Low           0
Close         0
Adj Close     0
Volume        0
dtype: int64
```

```
In [16]: meta.isnull().sum()
```

```
Out[16]: Date          0
Open          0
```

High 0
Low 0
Close 0
Adj Close 0
Volume 0
dtype: int64

In [17]:

microsoft.isnull().sum()

Out[17]: Date 0
Open 0
High 0
Low 0
Close 0
Adj Close 0
Volume 0
dtype: int64

In [18]:

netflix.isnull().sum()

Out[18]: Date 0
Open 0
High 0
Low 0
Close 0
Adj Close 0
Volume 0
dtype: int64

In [19]:

tesla.isnull().sum()

Out[19]: Date 0
Open 0
High 0
Low 0
Close 0
Adj Close 0
Volume 0
dtype: int64

Step 4: Adding Additional Columns to Each Dataset

In [20]:

```
# Inserting column to each dataset where it points out the stock name
amazon.insert(0, 'Stock_Name', "Amazon")
amazon
```

Out[20]:

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Volume
0	Amazon	2015-01-02	15.629000	15.737500	15.348000	15.426000	15.426000	556640
1	Amazon	2015-01-05	15.350500	15.419000	15.042500	15.109500	15.109500	554840
2	Amazon	2015-01-06	15.112000	15.150000	14.619000	14.764500	14.764500	703800

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Volu
3	Amazon	2015-01-07	14.875000	15.064000	14.766500	14.921000	14.921000	528060
4	Amazon	2015-01-08	15.016000	15.157000	14.805500	15.023000	15.023000	617680
...	...	...	...	...	...	...	...	...
2511	Amazon	2024-12-24	226.940002	229.139999	226.130005	229.050003	229.050003	150075
2512	Amazon	2024-12-26	228.500000	228.500000	226.669998	227.050003	227.050003	161465
2513	Amazon	2024-12-27	225.600006	226.029999	220.899994	223.750000	223.750000	273675
2514	Amazon	2024-12-30	220.059998	223.000000	218.429993	221.300003	221.300003	283215
2515	Amazon	2024-12-31	222.970001	223.229996	218.940002	219.389999	219.389999	248195

2516 rows x 8 columns

```
In [21]: apple.insert(0, 'Stock_Name', "Apple")
apple
```

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Vol
0	Apple	2015-01-02	27.847500	27.860001	26.837500	27.332500	24.347174	212818
1	Apple	2015-01-05	27.072500	27.162500	26.352501	26.562500	23.661274	257142
2	Apple	2015-01-06	26.635000	26.857500	26.157499	26.565001	23.663502	263188
3	Apple	2015-01-07	26.799999	27.049999	26.674999	26.937500	23.995319	160423
4	Apple	2015-01-08	27.307501	28.037500	27.174999	27.972500	24.917271	237458
...	...	...	...	...	...	...	...	...
2511	Apple	2024-12-24	255.490005	258.209991	255.289993	258.200012	258.200012	232345
2512	Apple	2024-12-26	258.190002	260.100006	257.630005	259.019989	259.019989	272345
2513	Apple	2024-12-27	257.829987	258.700012	253.059998	255.589996	255.589996	423567
2514	Apple	2024-12-30	252.229996	253.500000	250.750000	252.199997	252.199997	355567
2515	Apple	2024-12-31	252.440002	253.279999	249.429993	250.419998	250.419998	394805

2516 rows x 8 columns

In [22]:

```
meta.insert(0, 'Stock_Name', "Meta")
meta
```

Out[22]:

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Volu
0	Meta	2015-01-02	78.580002	78.930000	77.699997	78.449997	78.151466	18177
1	Meta	2015-01-05	77.980003	79.250000	76.860001	77.190002	76.896263	26452
2	Meta	2015-01-06	77.230003	77.589996	75.360001	76.150002	75.860222	27399
3	Meta	2015-01-07	76.760002	77.360001	75.820000	76.150002	75.860222	22045
4	Meta	2015-01-08	76.739998	78.230003	76.080002	78.180000	77.882500	23961
...	...	...	...	...	...	...	...	...
2511	Meta	2024-12-24	602.719971	607.989990	599.280029	607.750000	607.750000	4726
2512	Meta	2024-12-26	605.479980	606.299988	598.940002	603.349976	603.349976	6081
2513	Meta	2024-12-27	599.409973	601.849976	589.799988	599.809998	599.809998	8084
2514	Meta	2024-12-30	588.750000	596.940002	585.580017	591.239990	591.239990	7025
2515	Meta	2024-12-31	592.270020	593.969971	583.849976	585.510010	585.510010	6019

2516 rows x 8 columns

In [23]:

```
microsoft.insert(0, 'Stock_Name', "Microsoft")
microsoft
```

Out[23]:

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Volu
0	Microsoft	2015-01-02	46.660000	47.419998	46.540001	46.759998	40.152489	27913
1	Microsoft	2015-01-05	46.369999	46.730000	46.250000	46.330002	39.783257	39673
2	Microsoft	2015-01-06	46.380001	46.750000	45.540001	45.650002	39.199333	36447
3	Microsoft	2015-01-07	45.980000	46.459999	45.490002	46.230000	39.697376	29114
4	Microsoft	2015-01-08	46.750000	47.750000	46.720001	47.590000	40.865200	29645
...	...	...	...	...	...	...	...	...

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Volu
2511	Microsoft	2024-12-24	434.649994	439.600006	434.190002	439.329987	439.329987	7164
2512	Microsoft	2024-12-26	439.079987	440.940002	436.630005	438.109985	438.109985	8194
2513	Microsoft	2024-12-27	434.600006	435.220001	426.350006	430.529999	430.529999	18117
2514	Microsoft	2024-12-30	426.059998	427.549988	421.899994	424.829987	424.829987	13158
2515	Microsoft	2024-12-31	426.100006	426.730011	420.660004	421.500000	421.500000	13246

2516 rows x 8 columns

In [24]:

```
netflix.insert(0, 'Stock_Name', "Netflix")
netflix
```

Out[24]:

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Volu
0	Netflix	2015-01-02	49.151428	50.331429	48.731430	49.848572	49.848572	134750
1	Netflix	2015-01-05	49.258572	49.258572	47.147144	47.311428	47.311428	181650
2	Netflix	2015-01-06	47.347141	47.639999	45.661430	46.501431	46.501431	160370
3	Netflix	2015-01-07	47.347141	47.421429	46.271427	46.742859	46.742859	98490
4	Netflix	2015-01-08	47.119999	47.835712	46.478573	47.779999	47.779999	96010
...	...	...	...	...	...	...	...	...
2511	Netflix	2024-12-24	915.000000	935.849976	911.700012	932.119995	932.119995	23200
2512	Netflix	2024-12-26	928.400024	930.489990	915.299988	924.140015	924.140015	23400
2513	Netflix	2024-12-27	916.010010	918.130005	894.500000	907.549988	907.549988	32260
2514	Netflix	2024-12-30	894.510010	908.229980	889.710022	900.429993	900.429993	22030
2515	Netflix	2024-12-31	901.799988	902.679993	889.469971	891.320007	891.320007	18750

2516 rows x 8 columns

In [25]:

```
tesla.insert(0, 'Stock_Name', "Tesla")
tesla
```

Out [25]:

	Stock_Name	Date	Open	High	Low	Close	Adj Close	Vol
0	Tesla	2015-01-02	14.858000	14.883333	14.217333	14.620667	14.620667	71466
1	Tesla	2015-01-05	14.303333	14.433333	13.810667	14.006000	14.006000	80527
2	Tesla	2015-01-06	14.004000	14.280000	13.614000	14.085333	14.085333	93928
3	Tesla	2015-01-07	14.223333	14.318667	13.985333	14.063333	14.063333	44526
4	Tesla	2015-01-08	14.187333	14.253333	14.000667	14.041333	14.041333	51637
...	...	...	...	...	...	...	...	...
2511	Tesla	2024-12-24	435.899994	462.779999	435.140015	462.279999	462.279999	59551
2512	Tesla	2024-12-26	465.160004	465.329987	451.019989	454.130005	454.130005	76366
2513	Tesla	2024-12-27	449.519989	450.000000	426.500000	431.660004	431.660004	82666
2514	Tesla	2024-12-30	419.399994	427.000000	415.750000	417.410004	417.410004	64941
2515	Tesla	2024-12-31	423.790009	427.929993	402.540009	403.839996	403.839996	76825

2516 rows × 8 columns

Step 5: Converting values of columns to appropriate datatype

In [26]:

```
# Convert the datatype of Date column from object to Datetime.
amazon['Date'] = pd.to_datetime(amazon['Date'])
apple['Date'] = pd.to_datetime(apple['Date'])
meta['Date'] = pd.to_datetime(meta['Date'])
microsoft['Date'] = pd.to_datetime(microsoft['Date'])
netflix['Date'] = pd.to_datetime(netflix['Date'])
tesla['Date'] = pd.to_datetime(tesla['Date'])
```

In [27]:

```
# Convert the datatype of columns from object to float type.
amazon['Close in $'] = amazon['Close'].astype(float).round(4)
apple['Close in $'] = apple['Close'].astype(float).round(4)
meta['Close in $'] = meta['Close'].astype(float).round(4)
microsoft['Close in $'] = microsoft['Close'].astype(float).round(4)
netflix['Close in $'] = netflix['Close'].astype(float).round(4)
tesla['Close in $'] = tesla['Close'].astype(float).round(4)
```

In [28]:

```
amazon['Open in $'] = amazon['Open'].astype(float).round(4)
apple['Open in $'] = apple['Open'].astype(float).round(4)
meta['Open in $'] = meta['Open'].astype(float).round(4)
microsoft['Open in $'] = microsoft['Open'].astype(float).round(4)
netflix['Open in $'] = netflix['Open'].astype(float).round(4)
tesla['Open in $'] = tesla['Open'].astype(float).round(4)
```

```
In [29]: amazon['High in $'] = amazon['High'].astype(float).round(4)
apple['High in $'] = apple['High'].astype(float).round(4)
meta['High in $'] = meta['High'].astype(float).round(4)
microsoft['High in $'] = microsoft['High'].astype(float).round(4)
netflix['High in $'] = netflix['High'].astype(float).round(4)
tesla['High in $'] = tesla['High'].astype(float).round(4)
```

```
In [30]: amazon['Low in $'] = amazon['Low'].astype(float).round(4)
apple['Low in $'] = apple['Low'].astype(float).round(4)
meta['Low in $'] = meta['Low'].astype(float).round(4)
microsoft['Low in $'] = microsoft['Low'].astype(float).round(4)
netflix['Low in $'] = netflix['Low'].astype(float).round(4)
tesla['Low in $'] = tesla['Low'].astype(float).round(4)
```

```
In [31]: #To prep for moving averages and rolling over function, sort date column chronically
amazon.sort_values(by=['Date'], inplace=True)
apple.sort_values(by=['Date'], inplace=True)
meta.sort_values(by=['Date'], inplace=True)
microsoft.sort_values(by=['Date'], inplace=True)
netflix.sort_values(by=['Date'], inplace=True)
tesla.sort_values(by=['Date'], inplace=True)
```

```
In [32]: # Dropping the Old Columns: Close/Last, Open, High, Low
amazon.drop(columns = ['Close', 'Open', 'High', 'Low'], inplace = True)
apple.drop(columns = ['Close', 'Open', 'High', 'Low'], inplace = True)
meta.drop(columns = ['Close', 'Open', 'High', 'Low'], inplace = True)
microsoft.drop(columns = ['Close', 'Open', 'High', 'Low'], inplace = True)
netflix.drop(columns = ['Close', 'Open', 'High', 'Low'], inplace = True)
tesla.drop(columns = ['Close', 'Open', 'High', 'Low'], inplace = True)
```

```
In [33]: # Inserting columns to each dataset where it points out the 6 month and one year
```

```
In [34]: amazon['MA_Six_Months'] = amazon['Close in $'].rolling(window=183).mean()
amazon['MA_One_Year'] = amazon['Close in $'].rolling(window=365).mean()
amazon
```

```
Out[34]:
```

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_
0	Amazon	2015-01-02	15.426000	55664000	15.4260	15.6290	15.7375	15.3480	
1	Amazon	2015-01-05	15.109500	55484000	15.1095	15.3505	15.4190	15.0425	
2	Amazon	2015-01-06	14.764500	70380000	14.7645	15.1120	15.1500	14.6190	
3	Amazon	2015-01-07	14.921000	52806000	14.9210	14.8750	15.0640	14.7665	

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_1
4	Amazon	2015-01-08	15.023000	61768000	15.0230	15.0160	15.1570	14.8055	
...	...	...	...	...	...	...	...	...	
2511	Amazon	2024-12-24	229.050003	15007500	229.0500	226.9400	229.1400	226.1300	
2512	Amazon	2024-12-26	227.050003	16146700	227.0500	228.5000	228.5000	226.6700	
2513	Amazon	2024-12-27	223.750000	27367100	223.7500	225.6000	226.0300	220.9000	
2514	Amazon	2024-12-30	221.300003	28321200	221.3000	220.0600	223.0000	218.4300	
2515	Amazon	2024-12-31	219.389999	24819700	219.3900	222.9700	223.2300	218.9400	

2516 rows x 10 columns

In [35]:

```
apple['MA_Six_Months'] = apple['Close in $'].rolling(window=183).mean()  
apple['MA_One_Year'] = apple['Close in $'].rolling(window=365).mean()  
apple
```

Out[35]:

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_1
0	Apple	2015-01-02	24.347174	212818400	27.3325	27.8475	27.8600	26.8375	
1	Apple	2015-01-05	23.661274	257142000	26.5625	27.0725	27.1625	26.3525	
2	Apple	2015-01-06	23.663502	263188400	26.5650	26.6350	26.8575	26.1575	
3	Apple	2015-01-07	23.995319	160423600	26.9375	26.8000	27.0500	26.6750	
4	Apple	2015-01-08	24.917271	237458000	27.9725	27.3075	28.0375	27.1750	
...	...	...	...	...	...	...	...	...	
2511	Apple	2024-12-24	258.200012	23234700	258.2000	255.4900	258.2100	255.2900	
2512	Apple	2024-12-26	259.019989	27237100	259.0200	258.1900	260.1000	257.6300	
2513	Apple	2024-12-27	255.589996	42355300	255.5900	257.8300	258.7000	253.0600	
2514	Apple	2024-12-30	252.199997	35557500	252.2000	252.2300	253.5000	250.7500	
2515	Apple	2024-12-31	250.419998	39480700	250.4200	252.4400	253.2800	249.4300	

2516 rows x 10 columns

In [36]:

```
meta['MA_Six_Months'] = meta['Close in $'].rolling(window=183).mean()  
meta['MA_One_Year'] = meta['Close in $'].rolling(window=365).mean()  
meta
```

Out[36]:

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months
0	Meta	2015-01-02	78.151466	18177500	78.45	78.58	78.93	77.70	NaN
1	Meta	2015-01-05	76.896263	26452200	77.19	77.98	79.25	76.86	NaN
2	Meta	2015-01-06	75.860222	27399300	76.15	77.23	77.59	75.36	NaN
3	Meta	2015-01-07	75.860222	22045300	76.15	76.76	77.36	75.82	NaN
4	Meta	2015-01-08	77.882500	23961000	78.18	76.74	78.23	76.08	NaN
...	...	...	...	...	...	...	...	...	...
2511	Meta	2024-12-24	607.750000	4726100	607.75	602.72	607.99	599.28	528.60694
2512	Meta	2024-12-26	603.349976	6081400	603.35	605.48	606.30	598.94	529.02229
2513	Meta	2024-12-27	599.809998	8084200	599.81	599.41	601.85	589.80	529.4625
2514	Meta	2024-12-30	591.239990	7025900	591.24	588.75	596.94	585.58	529.86874
2515	Meta	2024-12-31	585.510010	6019500	585.51	592.27	593.97	583.85	530.22761

2516 rows x 10 columns

In [37]:

```
microsoft['MA_Six_Months'] = microsoft['Close in $'].rolling(window=183).mean()  
microsoft['MA_One_Year'] = microsoft['Close in $'].rolling(window=365).mean()  
microsoft
```

Out[37]:

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months
0	Microsoft	2015-01-02	40.152489	27913900	46.76	46.66	47.42	46.54	NaN
1	Microsoft	2015-01-05	39.783257	39673900	46.33	46.37	46.73	46.25	NaN
2	Microsoft	2015-01-06	39.199333	36447900	45.65	46.38	46.75	45.54	NaN
3	Microsoft	2015-01-07	39.697376	29114100	46.23	45.98	46.46	45.49	NaN

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Month
4	Microsoft	2015-01-08	40.865200	29645200	47.59	46.75	47.75	46.72	Na
...	...	...	...	...	...	...	...	...	...
2511	Microsoft	2024-12-24	439.329987	7164500	439.33	434.65	439.60	434.19	425.27469
2512	Microsoft	2024-12-26	438.109985	8194200	438.11	439.08	440.94	436.63	425.34349
2513	Microsoft	2024-12-27	430.529999	18117700	430.53	434.60	435.22	426.35	425.37599
2514	Microsoft	2024-12-30	424.829987	13158700	424.83	426.06	427.55	421.90	425.36809
2515	Microsoft	2024-12-31	421.500000	13246500	421.50	426.10	426.73	420.66	425.35849

2516 rows x 10 columns

In [38]:

```
netflix['MA_Six_Months'] = netflix['Close in $'].rolling(window=183).mean()  
netflix['MA_One_Year'] = netflix['Close in $'].rolling(window=365).mean()  
netflix
```

Out[38]:

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_
0	Netflix	2015-01-02	49.848572	13475000	49.8486	49.1514	50.3314	48.7314	
1	Netflix	2015-01-05	47.311428	18165000	47.3114	49.2586	49.2586	47.1471	
2	Netflix	2015-01-06	46.501431	16037700	46.5014	47.3471	47.6400	45.6614	
3	Netflix	2015-01-07	46.742859	9849700	46.7429	47.3471	47.4214	46.2714	
4	Netflix	2015-01-08	47.779999	9601900	47.7800	47.1200	47.8357	46.4786	
...	...	...	...	...	...	...	...	...	...
2511	Netflix	2024-12-24	932.119995	2320300	932.1200	915.0000	935.8500	911.7000	
2512	Netflix	2024-12-26	924.140015	2340300	924.1400	928.4000	930.4900	915.3000	
2513	Netflix	2024-12-27	907.549988	3226200	907.5500	916.0100	918.1300	894.5000	
2514	Netflix	2024-12-30	900.429993	2203000	900.4300	894.5100	908.2300	889.7100	
2515	Netflix	2024-12-31	891.320007	1875900	891.3200	901.8000	902.6800	889.4700	

2516 rows x 10 columns

In [39]:

```
tesla['MA_Six_Months'] = tesla['Close in $'].rolling(window=183).mean()  
tesla['MA_One_Year'] = tesla['Close in $'].rolling(window=365).mean()  
tesla
```

Out[39]:

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_
0	Tesla	2015-01-02	14.620667	71466000	14.6207	14.8580	14.8833	14.2173	
1	Tesla	2015-01-05	14.006000	80527500	14.0060	14.3033	14.4333	13.8107	
2	Tesla	2015-01-06	14.085333	93928500	14.0853	14.0040	14.2800	13.6140	
3	Tesla	2015-01-07	14.063333	44526000	14.0633	14.2233	14.3187	13.9853	
4	Tesla	2015-01-08	14.041333	51637500	14.0413	14.1873	14.2533	14.0007	
...	...	...	...	...	...	...	...	...	...
2511	Tesla	2024-12-24	462.279999	59551800	462.2800	435.9000	462.7800	435.1400	
2512	Tesla	2024-12-26	454.130005	76366400	454.1300	465.1600	465.3300	451.0200	
2513	Tesla	2024-12-27	431.660004	82666800	431.6600	449.5200	450.0000	426.5000	
2514	Tesla	2024-12-30	417.410004	64941000	417.4100	419.4000	427.0000	415.7500	
2515	Tesla	2024-12-31	403.839996	76825100	403.8400	423.7900	427.9300	402.5400	

2516 rows x 10 columns

Step 6: Checking For Duplicates

In [40]:

```
amazon[amazon.duplicated()]
```

Out[40]:

Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
------------	------	-----------	--------	-------------	------------	------------	-----------	---------------	-------------

In [41]:

```
apple[apple.duplicated()]
```

Out[41]:

Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
------------	------	-----------	--------	-------------	------------	------------	-----------	---------------	-------------

In [42]:

```
meta[meta.duplicated()]
```

Out[42]:

Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
------------	------	-----------	--------	-------------	------------	------------	-----------	---------------	-------------

In [43]:

```
microsoft[microsoft.duplicated()]
```

Out[43]:

Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
------------	------	-----------	--------	-------------	------------	------------	-----------	---------------	-------------

In [44]:

```
netflix[netflix.duplicated()]
```

Out[44]:

Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
------------	------	-----------	--------	-------------	------------	------------	-----------	---------------	-------------

In [45]:

```
tesla[tesla.duplicated()]
```

Out[45]:

Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months	MA_One_Year
------------	------	-----------	--------	-------------	------------	------------	-----------	---------------	-------------

Step 7: Combine All Datasets

In [46]:

```
frames_stocks = [amazon,apple,meta,microsoft,netflix,tesla]

stocks = pd.concat(frames_stocks, axis = 0)
stocks
```

Out[46]:

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_
0	Amazon	2015-01-02	15.426000	55664000	15.4260	15.6290	15.7375	15.3480	
1	Amazon	2015-01-05	15.109500	55484000	15.1095	15.3505	15.4190	15.0425	
2	Amazon	2015-01-06	14.764500	70380000	14.7645	15.1120	15.1500	14.6190	
3	Amazon	2015-01-07	14.921000	52806000	14.9210	14.8750	15.0640	14.7665	
4	Amazon	2015-01-08	15.023000	61768000	15.0230	15.0160	15.1570	14.8055	
...	...	...	...	...	...	...	...	...	...
2511	Tesla	2024-12-24	462.279999	59551800	462.2800	435.9000	462.7800	435.1400	
2512	Tesla	2024-12-26	454.130005	76366400	454.1300	465.1600	465.3300	451.0200	
2513	Tesla	2024-12-27	431.660004	82666800	431.6600	449.5200	450.0000	426.5000	

	Stock_Name	Date	Adj Close	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_
2514	Tesla	2024-12-30	417.410004	64941000	417.4100	419.4000	427.0000	415.7500	
2515	Tesla	2024-12-31	403.839996	76825100	403.8400	423.7900	427.9300	402.5400	

15096 rows × 10 columns

In [47]:

stocks.dtypes

Out[47]:

Stock_Name	object
Date	datetime64[ns]
Adj Close	float64
Volume	int64
Close in \$	float64
Open in \$	float64
High in \$	float64
Low in \$	float64
MA_Six_Months	float64
MA_One_Year	float64
dtype:	object

In [48]:

```
# Reordering Columns
stocks = stocks[['Date', 'Stock_Name', 'Volume', 'Close in $', 'Open in $', 'High in $', 'Low in $', 'MA_Six_Months']]
```

Out[48]:

	Date	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months
0	2015-01-02	Amazon	55664000	15.4260	15.6290	15.7375	15.3480	NaN
1	2015-01-05	Amazon	55484000	15.1095	15.3505	15.4190	15.0425	NaN
2	2015-01-06	Amazon	70380000	14.7645	15.1120	15.1500	14.6190	NaN
3	2015-01-07	Amazon	52806000	14.9210	14.8750	15.0640	14.7665	NaN
4	2015-01-08	Amazon	61768000	15.0230	15.0160	15.1570	14.8055	NaN
...	...	...	...	...	...	...	...	...
2511	2024-12-24	Tesla	59551800	462.2800	435.9000	462.7800	435.1400	239.391967
2512	2024-12-26	Tesla	76366400	454.1300	465.1600	465.3300	451.0200	240.972459
2513	2024-12-27	Tesla	82666800	431.6600	449.5200	450.0000	426.5000	242.386011
2514	2024-12-30	Tesla	64941000	417.4100	419.4000	427.0000	415.7500	243.700383

	Date	Stock_Name	Volume	Close in \$	Open in \$	High in \$	Low in \$	MA_Six_Months
2515	2024-12-31	Tesla	76825100	403.8400	423.7900	427.9300	402.5400	244.968579

15096 rows x 9 columns

Step 8: Exporting drafted dataframe

In [49]:

```
# Exporting drafted dataframe
stocks.to_csv(os.path.join(path, 'New Files', '6_Stocks_Transformed.csv'))
```